

Thiago de Paula Oliveira

Statistician | Agri-Tech Data Analytics | AI Integration | Reproducible Workflows | Technology Evaluation

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EXECUTIVE PROFILE

Statistician and analytical software developer with 14+ years of experience applying statistical modelling, reproducible data workflows, and decision-support tools to agriculture, quantitative genetics, breeding, trial analysis, and applied monitoring problems. Experienced in converting complex datasets into robust evidence, technical reports, dashboards, and software workflows that support scientific, operational, and commercial decision-making. Now seeking to apply this background to agri-tech data, AI-enabled workflows, and technology-evaluation programmes.

CAREER HIGHLIGHTS

- **Delivers agri-tech evidence and decision-support workflows** across livestock, crop, genomic-QC, trial-analysis, genetic-evaluation, and sustainability-index programmes.
- **Builds reproducible analytical systems** in R, C++, SQL, Bash, Shiny, Docker, and Git, with emphasis on validation, transparent data processing, reporting, and handover-ready outputs.
- **Supports technology evaluation and benchmarking** by translating complex modelling outputs into defensible evidence, technical reports, dashboards, and stakeholder-ready recommendations.
- **Bridges agricultural domain knowledge and statistical computing** through experience in agricultural engineering, quantitative genetics, breeding analytics, sustainability metrics, and applied monitoring.
- **Uses LLM/LRMs carefully within analytical software workflows** for implementation planning, refactoring, documentation, code review, testing strategy, and reproducible delivery.

CORE COMPETENCIES AND ROLE RELEVANCE

STATISTICAL METHODS	Mixed models, Bayesian analysis, multilevel modelling, longitudinal analysis, simulation, experimental design
DATA AND SOFTWARE	R, C++, SQL, Bash, Shiny, Docker, CI, Git, reproducible pipelines, package development
AI-ASSISTED DELIVERY	Implementation planning, code review, refactoring, documentation, test planning, validation, quality-controlled analytical workflows
AGRI-TECH DOMAIN	Quantitative genetics, breeding-program analytics, genomic QC, economic/sustainability indices, trial analysis, applied agricultural experimentation
EVIDENCE GENERATION	Decision-support tooling, benchmarking, technical reporting, dashboards, stakeholder-ready recommendations
STAKEHOLDER DELIVERY	Communication with scientists, breeders, commercial teams, clinicians, and applied-domain specialists; mentoring and code review

PROFESSIONAL EXPERIENCE

APR 2023 – PRESENT | **Consultant Statistician**
[AbacusBio](#)

- Lead statistical and analytical delivery for livestock, crop, and agri-tech programmes, developing decision-support workflows for genetic evaluation, genomic QC, trial analysis, and technology-focused evidence generation.

- Design economic and sustainability selection indices that translate biological, operational, environmental, and commercial objectives into transparent ranking and reporting frameworks.
- Build reproducible analytical workflows in R, C++, SQL, and Bash, using Docker and Git to improve data processing, validation, reproducibility, handover quality, and deployment readiness.
- Develop and validate statistical software workflows, including implementation briefs, code refactoring, documentation, test planning, and checking outputs against expected statistical behaviour.
- Deliver Shiny applications, technical reports, and stakeholder summaries that convert complex statistical outputs into practical recommendations for scientists, breeders, and commercial teams.
- Mentor colleagues through code review, modelling guidance, package-development standards, reproducible workflows, and appropriate use of AI-assisted analytical development.

OCT 2021 – MAR 2023 | **Research Fellow in Quantitative Genetics**
[The Roslin Institute, University of Edinburgh](#)

- Awarded a Marie Skłodowska-Curie COFUND fellowship focused on quantitative-genetic methods for sustainability-oriented breeding programmes.
- Developed mixed-model, Bayesian, and simulation approaches for plant-breeding data, selection strategy, and programme monitoring.
- Produced methods and analytical workflows for partitioning trends in genetic mean and additive genetic variance.
- Collaborated with genetics, breeding, and computing teams to translate methodology into implementable analytical pipelines.

APR 2019 – OCT 2021 | **Postdoctoral Researcher in Statistics**
[Insight Centre for Data Analytics, University of Galway](#)

- Developed Bayesian, multilevel, and longitudinal models for athlete readiness, fatigue, menstrual-cycle dynamics, and health-monitoring programmes.
- Built forecasting and dashboard pipelines for COVID-19 and related public-health decision support.
- Worked with clinicians and sports scientists to convert modelling outputs into practical tools and reporting workflows.

2019 | **Statistical Consultant**
[Orreco](#)

- Developed statistical models and dashboard outputs for elite-athlete monitoring, recovery, and performance support using biomarker, GPS, and wellness data.

2017 – 2019 | **Lecturer in Statistics**
[University of São Paulo \(ESALQ/USP\)](#)

- Taught experimental design, calculus, linear models, and statistical programming at undergraduate and postgraduate levels.
- Supervised applied projects connecting statistics with agriculture and genetics, with emphasis on reproducible workflows.

■ ANALYTICAL SOFTWARE AND OPEN-SOURCE CONTRIBUTIONS

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- **matrixCorr** — R package for correlation and agreement estimation in a unified framework for method comparison and applied reporting. [[GitHub](#); [CRAN](#)]
 - **lcc** — CRAN package for longitudinal concordance and accuracy profiling using polynomial mixed-effects models. [[GitHub](#); [CRAN](#)]
 - **Selected contributions** — Contributions to [AlphaSimR](#), [AGHmatrix](#), [AlphaPart](#), and [Reads2Map Tools](#) in breeding simulation, matrix methods, and genotyping workflows. [[GitHub](#); [GitHub](#); [GitHub](#); [GitHub](#)]

■ EDUCATION

- 2014 – 2018 | **PhD in Statistics**
[University of São Paulo \(ESALQ/USP\)](#)
Dissertation on longitudinal concordance analysis using polynomial mixed-effects models.
Advisors: Silvio Sandoval Zocchi and John Hinde. [[PDF](#)]
- 2016 | **Visiting Research Internship**
[University of Galway](#)
Research placement in concordance-analysis methodology under the supervision of John Hinde.
- 2012 – 2014 | **MSc in Statistics**
[University of São Paulo \(ESALQ/USP\)](#)
Research in mixed-effects models for longitudinal fruit-colour measurement.
- 2007 – 2012 | **BSc in Agricultural Engineering**
[University of São Paulo \(ESALQ/USP\)](#)
Training in agricultural systems, experimentation, and quantitative analysis.

■ SELECTED PUBLICATIONS

- Richardson, C., Amer, P., Post, M., **Oliveira, T. d. P.**, et al. (2025). [Breeding for sustainability: development of an index to reduce greenhouse gas in dairy cattle](#). *Animal*.
- Oliveira, T. d. P.** & Newell, J. (2024). [A hierarchical approach for evaluating athlete performance with an application in elite basketball](#). *Scientific Reports*.
- Oliveira, T. d. P.**, Obsteter, J., Pocrnic, I., Heslot, N. & Gorjanc, G. (2023). [A method for partitioning trends in genetic mean and variance to understand breeding practices](#). *Genetics Selection Evolution*.
- Oliveira, T. d. P.**, Bruinvels, G., Pedlar, C., Moore, B. & Newell, J. (2021). [Modelling menstrual cycle length in athletes using state-space models](#). *Scientific Reports*.

■ AWARDS

- 2020 — [Marie Skłodowska-Curie COFUND Fellowship](#), quantitative genetics and genomics of plant breeding.
- 2010 — Honorable Mention, 18th USP International Symposium of Undergraduate Research, University of São Paulo.

■ LANGUAGES

- Portuguese (native), English (professional proficiency), Spanish (reading/listening).